

1) $\int \frac{\cos x \sin x}{1 + \sin^4 x} dx$ $t = \sin^2 x \quad \frac{dt}{dx} = 2 \sin x \cos x$
 原式 = $\int \frac{\cos x \sin x}{1 + t^2} \cdot \frac{dt}{2 \sin x \cos x} = \int \frac{1}{2} \frac{dt}{1 + t^2} = \frac{1}{2} \arctan t + C$

2) $\int \arctan \sqrt{x} dx = x \arctan \sqrt{x} - \int \frac{x}{1+x} \cdot \frac{1}{2\sqrt{x}} dx$
 $\frac{1}{2} \int \frac{\sqrt{x}}{1+x} dx \quad t = \sqrt{x} \quad \frac{1}{2} \int \frac{t}{1+t^2} \cdot 2t dt = \int \frac{t^2}{1+t^2} dt = \int (1 - \frac{1}{1+t^2}) dt = t - \arctan t + C$
 $\frac{dt}{dx} = \frac{1}{2\sqrt{x}} \Rightarrow \frac{1}{2} \sqrt{x} = \frac{1}{2} \frac{1}{\sqrt{x}}$
 数学分析 II (2022-2023 学年) $\int \sqrt{x} - \arctan \sqrt{x} + C$

一. (18 分) 计算下列积分: (3) $\int_0^{\frac{\pi}{4}} \frac{\sin x}{\sin x + \cos x} dx$ $t = \tan \frac{x}{2} \quad \sin x = \frac{2 \sin \frac{x}{2} \cos \frac{x}{2}}{\sin^2 \frac{x}{2} + \cos^2 \frac{x}{2}} = \frac{2t}{t^2+1}$ $\cos x = \frac{1-t^2}{t^2+1}$ $\frac{dx}{dt} = \frac{2}{t^2+1}$

(1) $\int \frac{\cos x \sin x}{1 + \sin^4 x} dx$; (2) $\int \arctan \sqrt{x} dx$; (3) $\int_0^{\frac{\pi}{4}} \frac{\sin x}{\sin x + \cos x} dx$
 原式 = $\int_0^{\frac{\pi}{4}} \frac{2t}{t^2+1} \cdot \frac{2dt}{t^2+1} = \int_0^{\frac{\pi}{4}} \frac{4t}{(t^2+1)^2} dt$

二. (10 分) 设

$a_n = \frac{1}{n} \sqrt{(n+1)(n+2) \dots (n+n)}, n \in \mathbb{Z}^+$ $-\frac{1}{2} \ln |t-1| + \frac{1}{2} \ln |t+1| \Big|_0^{\frac{\sqrt{3}}{2}} - \arctan t \Big|_0^{\frac{\sqrt{3}}{2}} = \dots$

求 $\lim_{n \rightarrow \infty} a_n$ $\lim_{n \rightarrow \infty} \frac{\sqrt{(n+1)(n+2) \dots (n+n)}}{n} = \lim_{n \rightarrow \infty} \sqrt{(1+\frac{1}{n})(1+\frac{2}{n}) \dots (1+\frac{n}{n})} = T$ $-\frac{1}{2} \ln(t^2+1) \Big|_0^{\frac{\sqrt{3}}{2}} - \arctan t \Big|_0^{\frac{\sqrt{3}}{2}} = \dots$

$\ln T = \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n \ln(1 + \frac{k}{n})$ (p) $1 = x_0 < x_1 < x_2 < \dots < x_n = 2$ $x_i = 1 + \frac{i}{n}$

三. (12 分) 设函数 f 在区间 $[a, b]$ 上连续, 且满足 $\ln T = \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n \ln f(x_k) = \int_1^2 \ln x dx = (x \ln x - x) \Big|_1^2 = 2 \ln 2 - 1$
 $\int_a^b x^i f(x) dx = 0, (i = 0, 1, \dots, n)$ $f(x) = \ln x$

证明: f 在区间 (a, b) 内至少有两个零点.

$\int_a^b x^i f(x) dx = 0 \quad (i=0, 1, \dots, n)$ $\int_a^b f(x) dx = 0$ $\int_a^b x f(x) dx = 0$ $\Rightarrow \exists \xi \in (a, b)$ s.t. $\int_a^{\xi} f(x) dx = 0$ $\Rightarrow f$ 不恒为 0

四. (12 分) 设 f 是区间 $[a, b]$ 上的非负可积函数, 有有限个间断点. 证明:

$\int_a^b f(x) dx = 0$ $\Rightarrow f(x) = 0$ a.e. $\Rightarrow f(x) = 0$ 几乎处处成立

仅当 f 在连续点上恒为 0. $\int_a^b f(x) dx = 0 \Rightarrow \int_a^b f(x) dx = 0$

$\Rightarrow$ 局部保号性 $\int_a^b f(x) dx = 0, f(x) \geq 0$ 证明: 假设 $\exists x_0$ 为连续点, 且 $f(x_0) > 0 \Rightarrow f(x) > 0$

取 $\varepsilon = \frac{f(x_0)}{2} > 0$ 则 $\exists \delta > 0$ s.t. 连续点 $x \in U(x_0, \delta)$ 时, 有 $f(x) > \varepsilon$
 $\therefore \int_{x_0-\delta}^{x_0+\delta} f(x) dx > \int_{x_0-\delta}^{x_0+\delta} \varepsilon dx = 2\delta \varepsilon > 0$ 矛盾!

五. (12 分) 设 f 是 $\mathbb{R}$ 上的连续函数. 已知 $\int_0^1 f(tx) dt = 0$ 对任意 $x \in \mathbb{R}$ 成立 $f(x) \equiv 0$.

$\int_0^1 f(tx) dt = 0 \quad x=0: \int_0^1 f(0) dt = 0 \Rightarrow f(0) = 0$
 $x \neq 0$ 时 $\int_0^1 f(tx) dt = 0 \Rightarrow \frac{1}{x} F(x) \Big|_0^1 = 0 \Rightarrow F(x) - F(0) = 0$

六. (12 分) 求曲线 $r = \frac{1}{1+\cos \theta}, \theta \in [-\frac{\pi}{2}, \frac{\pi}{2}]$ 的弧长.

弧长: $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sqrt{r(\theta)^2 + r'(\theta)^2} d\theta = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sqrt{\frac{1}{(1+\cos \theta)^2} + \frac{\sin^2 \theta}{(1+\cos \theta)^4}} d\theta$

七. (12 分) 求位于“曲线 $r = 1 + 2 \cos \theta$ 以内, 圆 $r = 2$ 以外”这部分区域的面积.
 $= \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{1}{2} (r_1^2 - r_2^2) d\theta = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{1}{2} (\sin^2 \theta + \cos^2 \theta) d\theta = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{1}{2} d\theta = \frac{\pi}{2}$

八. (12 分) 设 $\alpha > 0$, 请讨论无穷积分 $\int_1^{\infty} \frac{\ln^{\alpha} x \sin x}{x} dx$ 的收敛性. 其中 $\int \frac{\sin t}{\cos^2 t} dt = -\int \frac{d \cos t}{\cos^2 t} = \frac{1}{\cos t} + C$

$\int \frac{1}{\cos t} dt = \int \frac{d \sin t}{1 - \sin^2 t} = \frac{1}{2} \ln \left| \frac{1 + \sin t}{1 - \sin t} \right| + C = \ln \left| \frac{1 + \sin t}{\cos t} \right| + C$
 $\int_1^{\infty} \frac{\ln^{\alpha} x \sin x}{x} dx$ $(\alpha > 0)$ $\therefore$ 函数连续 $\therefore \int_1^{\infty} \frac{\ln^{\alpha} x \sin x}{x} dx$ 为常函数 (解)

$\therefore$ 对任意 $\alpha > 0$, $\int_1^{\infty} \frac{\ln^{\alpha} x \sin x}{x} dx$ 收敛. $\therefore \frac{\ln^{\alpha} x}{x}$ 单调趋于 0 $\therefore$ 由 Dirichlet 判别法收敛.

$S = 2 \int_0^{\frac{\pi}{2}} \frac{1}{2} ((1+2\cos \theta)^2 - 2^2) d\theta$ 极坐标算面积: $S = \int_{\alpha}^{\beta} \frac{1}{2} r^2 d\theta$ $S = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{1}{2} (4 - 4 \cos^2 \theta) d\theta = \frac{\pi}{2}$

$$- \text{ts. (ii)} \int_0^{+\infty} \sin \frac{\ln x}{(x+\frac{1}{x})^\alpha} dx (\alpha > 0) = \int_0^1 \sin \frac{\ln x}{(x+\frac{1}{x})^\alpha} dx + \int_1^{+\infty} \sin \frac{\ln x}{(x+\frac{1}{x})^\alpha} dx$$

I. $x > 0, x + \frac{1}{x} \geq 2$ $(x + \frac{1}{x})^\alpha \geq 2^\alpha$ $I = \frac{\ln x}{(x + \frac{1}{x})^\alpha} \leq \frac{\ln x}{2^\alpha}$

$$\lim_{x \rightarrow 0} \frac{\ln x}{x^\alpha} = \lim_{x \rightarrow 0} \frac{-\frac{1}{x}}{\alpha x^{\alpha-1}} = \lim_{x \rightarrow 0} -\frac{1}{\alpha x^\alpha} = 0$$

$$\therefore \int_0^1 \sin \frac{\ln x}{(x + \frac{1}{x})^\alpha} dx \leq \int_0^1 \frac{1}{2^\alpha} dx = \frac{1}{2^\alpha}$$

II. $x > 1, x + \frac{1}{x} \leq x + \frac{1}{x} \leq 2x$ $(x + \frac{1}{x})^\alpha \leq (2x)^\alpha = 2^\alpha x^\alpha$

$$\lim_{x \rightarrow +\infty} \frac{\ln x}{x^\alpha} = \lim_{x \rightarrow +\infty} \frac{\frac{1}{x}}{\alpha x^{\alpha-1}} = \lim_{x \rightarrow +\infty} \frac{1}{\alpha x^\alpha} = 0$$

$$\therefore \int_1^{+\infty} \sin \frac{\ln x}{(x + \frac{1}{x})^\alpha} dx \leq \int_1^{+\infty} \frac{1}{2^\alpha x^\alpha} dx = \frac{1}{2^\alpha} \int_1^{+\infty} \frac{1}{x^\alpha} dx = \frac{1}{2^\alpha} \cdot \frac{1}{\alpha-1} \quad (\alpha > 1)$$

$$- \lim_{n \rightarrow \infty} n \left[\frac{1}{(n+1)^2} + \dots + \frac{1}{(2n)^2} \right]$$

$$\forall n \geq 1 \quad \frac{1}{(n+1)(n+2)} < \frac{1}{(n+1)^2} < \frac{1}{n(n+1)}$$

$$\text{累加得} \quad \frac{n^2}{2(n+1)^2} < \frac{n^2}{(n+1)(2n+1)} \leq \sqrt[n]{n!} \leq \frac{1}{2}$$

$$\forall n \text{ 充分大} \quad \lim_{n \rightarrow \infty} \frac{n^2}{2(n+1)^2} = \frac{1}{2} \quad \therefore \text{原式} \quad \sqrt[n]{n!} = \frac{1}{2}$$

数学分析 II

(2023-2024 学年)

一、(10 分) 求极限

$$\lim_{n \rightarrow \infty} n \left[\frac{1}{(n+1)^2} + \dots + \frac{1}{(2n)^2} \right]$$

二、(12 分) (i) 设 $F(x)$ 是 $f(x)$ 在 $(-1,1)$ 内的原函数且满足

$$f(x) = \frac{F(x)}{\sqrt{1-x^2}}, x \in (-1,1)$$

求 $f(x)$.

$$f(x) = F'(x) = \frac{F(x)}{\sqrt{1-x^2}} - \frac{dF(x)}{dx} \Rightarrow \frac{dx}{\sqrt{1-x^2}} = \frac{dF(x)}{F(x)} \Rightarrow \arcsin x + C = \ln|F(x)|$$

$$\text{则 } f(x) = \frac{C_1}{\sqrt{1-x^2}} e^{\arcsin x} \Rightarrow |f(x)| = e^{\arcsin x + C} = C_1 e^{\arcsin x} \quad (C_1 = e^C)$$

(ii) 求不定积分 $\int \frac{1}{x\sqrt{1-x^2}} dx$. $\frac{1}{2} t = \sqrt{1-x^2} \quad \frac{dt}{dx} = -\frac{x}{\sqrt{1-x^2}} \Rightarrow dx = -\frac{\sqrt{1-x^2}}{x} dt$

$$\text{原式} = \int \frac{1}{x\sqrt{1-x^2}} \cdot \frac{\sqrt{1-x^2}}{x} dt = \int \frac{dt}{x^2} = \int \frac{dt}{t^2 - 1} = \int \frac{1}{2} \left(\frac{1}{t-1} - \frac{1}{t+1} \right) dt = \frac{1}{2} \ln \left| \frac{t-1}{t+1} \right| + C$$

三、(14 分) 计算

(i) $\int_{-1}^1 (\tan x + \sqrt{1-x^2})^2 dx$

(ii) $\int_1^{+\infty} \frac{1}{x^2 + x^4 + 1} dx = \frac{1}{2} \int_1^{+\infty} \frac{1}{x^2 + x^4} \frac{dx}{x^2} = \frac{1}{2} \int_1^{+\infty} \frac{1}{1 + x^6} dx = \frac{1}{2} \arctan x \Big|_1^{+\infty} = \frac{\pi}{2} - \arctan 1 = \frac{\pi}{4}$

iv) $\int_{-1}^1 (\tan^2 x + 1 - x^2 + 2\sqrt{1-x^2} \tan x) dx = \int_{-1}^1 (\tan^2 x + 1) dx + \int_{-1}^1 (-x^2) dx + \int_{-1}^1 2\sqrt{1-x^2} \tan x dx$

四、(14 分) 求曲线: $r = 2(1 - \cos \theta), 0 \in [0, 2\pi)$ 的周长以及该曲线绕极轴旋转一周所得旋转曲面的表面积.

其中 $\int_{-1}^1 (\tan^2 x + 1) dx = \int_{-1}^1 \frac{1}{\cos^2 x} dx = \tan x \Big|_{-1}^1 = \frac{\pi}{2}$

$\int_{-1}^1 2\sqrt{1-x^2} \tan x dx$ $\because g(x) = 2\sqrt{1-x^2} \tan x$

$g(x) = 2\sqrt{1-x^2} \tan x = -2\sqrt{1-x^2} \tan x = -g(x)$ 为奇函数 $\therefore \int_{-1}^1 g(x) dx = 0$

五、(13 分) 设 $\frac{2}{3}$

原上原式 $= \frac{\pi}{2} - \frac{\pi}{3}$

$R(x) = \begin{cases} \frac{1}{q}, x = \frac{p}{q}, \text{其中 } p, q \text{ 为互质正整数且 } p < q \\ 0, x = 0, 1 \text{ 或 } (0,1) \text{ 内的无理数} \end{cases}$

证明: $R(x)$ 在 $[0,1]$ 上可积. $\forall \varepsilon > 0$ 满足 $R(x) \geq \varepsilon$ 的区间 I 的总长度 $< \frac{\varepsilon}{2}$. $\forall \varepsilon > 0$ 取 $p, q \geq \frac{1}{\varepsilon}$ 则 $\frac{1}{q} \geq \varepsilon \Rightarrow q \leq \frac{1}{\varepsilon}$, 即 $q = 2, 3, \dots, \frac{1}{\varepsilon}$, $\frac{1}{2} \leq \frac{1}{q} \leq 1$ 则 $\frac{1}{q} \geq \varepsilon$ 的区间 I 的总长度 $< \frac{\varepsilon}{2}$.

六、(12 分) 设 $f(x)$ 在 $[0,1]$ 上连续且单调递减, 证明: 对任意的 $\alpha \in [0,1]$, $\int_0^\alpha f(x) dx \geq \alpha \int_0^1 f(x) dx$

$\int_0^\alpha f(x) dx \geq \alpha \int_0^1 f(x) dx$ $\frac{1}{2} g(t) = \int_0^t f(x) dx - t \int_0^1 f(x) dx \quad t \in [0,1]$

$t=0$ 时 $g(0)=0$ $t=1$ 时 $g(1)=0$ $g'(t) = f(t) - \int_0^1 f(x) dx$

七、(16 分) 判断下列反常积分的敛散性 (请指出是绝对收敛、条件收敛还是发散)

(i) $\int_0^{+\infty} \sin \left(\frac{\ln x}{(x+\frac{1}{x})^\alpha} \right) dx (\alpha > 0)$ (ii) $\int_0^{+\infty} \frac{\sin x}{x^p + x^q} dx (p, q > 0)$

$\int_0^{+\infty} \sin \frac{\ln x}{(x+\frac{1}{x})^\alpha} dx (\alpha > 0)$ $\int_0^{+\infty} \frac{\sin x}{x^p + x^q} dx (p, q > 0)$

八、(10 分) 设 $m < 2$, 证明:

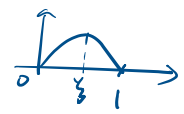
$$\lim_{x \rightarrow 0^+} x^{-m} \int_0^x \sin \frac{1}{t} dt = 0$$

由第一中值定理 $\exists \xi \in (0,1)$ 使 $f(\xi) = \int_0^1 f(x) dx$

$$\therefore g'(t) = f(t) - f(\xi) \quad \begin{cases} 0 < t < \xi \text{ 时 } g'(t) > 0 \\ \xi < t < 1 \text{ 时 } g'(t) < 0 \end{cases}$$

$\therefore g(t)$ 在 $(0,\xi)$ 上 $\uparrow$, 在 $(\xi,1)$ 上 $\downarrow$

$\forall t \in [0,1]$ 时, $g(t) \geq 0$ 证毕



12. $r = 2(1 - \cos \theta) \quad \theta \in [0, 2\pi)$ $r'(0) = 2 \sin 0 = 0$ $\cos \theta = 1 - 2 \sin^2 \frac{\theta}{2}$

$$L = \int_0^{2\pi} \sqrt{r(\theta)^2 + r'(\theta)^2} d\theta = \int_0^{2\pi} \sqrt{4 + 4 \sin^2 \frac{\theta}{2}} d\theta = 8 \int_0^{2\pi} \sin \frac{\theta}{2} d\frac{\theta}{2} = -8 \cos \frac{\theta}{2} \Big|_0^{2\pi} = 16$$

例面积 $S = 2\pi \int_0^{2\pi} g(\theta) \sqrt{r(\theta)^2 + g'(\theta)^2} dt$

$$= 2\pi \int_0^{2\pi} 2(1 - \cos \theta) \sin \theta \sqrt{4 + 4 \sin^2 \frac{\theta}{2}} d\theta$$

$$= 64\pi \int_0^{\frac{\pi}{2}} \sin^4 \frac{\theta}{2} \cos \frac{\theta}{2} d\frac{\theta}{2}$$

$$= 64\pi \int_0^{\frac{\pi}{2}} \sin^4 \frac{\theta}{2} \cos \frac{\theta}{2} d\frac{\theta}{2} = 128\pi \int_0^1 \sin^4 \frac{\theta}{2} d\sin \frac{\theta}{2} = \frac{128\pi}{5} \sin^5 \frac{\theta}{2} \Big|_0^1 = \frac{128\pi}{5}$$

$r = r(\theta)$ $\star x(0) = r(0) \cos 0 = 2(1 - \cos 0) = 0$ $x'(0) = 2(2 \sin 0 \cos 0 - \sin 0) = 0$

$y(0) = r(0) \sin 0 = 2(1 - \cos 0) \sin 0 = 0$ $y'(0) = 2(\sin 0 - \cos 0 + \sin 0) = 0$

$x'(0)^2 + y'(0)^2 = 8(1 - \cos 0) = 16 \sin^2 \frac{0}{2}$

ts. (ii) $\int_0^{+\infty} \frac{\sin x}{x^p + x^q} dx (p, q > 0)$ $= \int_0^1 \frac{\sin x}{x^p + x^q} dx + \int_1^{+\infty} \frac{\sin x}{x^p + x^q} dx$

$\forall x > 0$ 时 $\lim_{x \rightarrow 0} \frac{\sin x}{x^p + x^q} = \lim_{x \rightarrow 0} \frac{\cos x}{p x^{p-1} + q x^{q-1}} = \frac{1}{p+q}$ $\therefore x=0$ 不是奇点

$\int_0^1 \frac{\sin x}{x^p + x^q} dx < \int_0^1 \frac{1}{x^p + x^q} dx$ 若 p, q 中有一个 > 1 , $\int_0^1 \frac{1}{x^p + x^q} dx$ 收敛

$\int_1^{+\infty} \frac{\sin x}{x^p + x^q} dx < \int_1^{+\infty} \frac{1}{x^p + x^q} dx = \int_1^{+\infty} \frac{1}{x^{\min(p,q)}} dx = \frac{1}{\min(p,q)-1} \quad (\min(p,q) > 1)$

若 p, q 都 < 1 时 $\int_1^{+\infty} \frac{1}{x^p + x^q} dx$ 发散

1. $m < 2$ $\lim_{x \rightarrow 0^+} x^{-m} \int_0^x \sin \frac{1}{t} dt = 0$

证: $\int_0^x \sin \frac{1}{t} dt = \int_0^x t^2 \frac{\sin \frac{1}{t}}{t^2} dt = t^2 \cos \frac{1}{t} \Big|_0^x - \int_0^x 2t \cos \frac{1}{t} dt$

$\therefore \int_0^x \sin \frac{1}{t} dt = O(x^2)$

原式 $\lim_{x \rightarrow 0^+} x^{-m} x^2 = \lim_{x \rightarrow 0^+} x^{2-m} = 0 \quad (2-m > 0)$